TECHNOLOGY

Designing Microsoft Azure Infrastructure Solutions: AZ:305

TECHNOLOGY

Design Data Integration

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Learning Objectives

By the end of this lesson, you will be able to:

- Comprehend the use cases of Azure Data Services to optimize the data solutions effectively and make informed decisions
- Use Azure Data Lake Storage as a scalable and secure solution for big data analytics
- Gain proficiency in leveraging Azure Databricks for accelerating big data analytics and AI solutions
- Use Azure Stream Analytics effectively for analyzing and processing large volumes of streaming data

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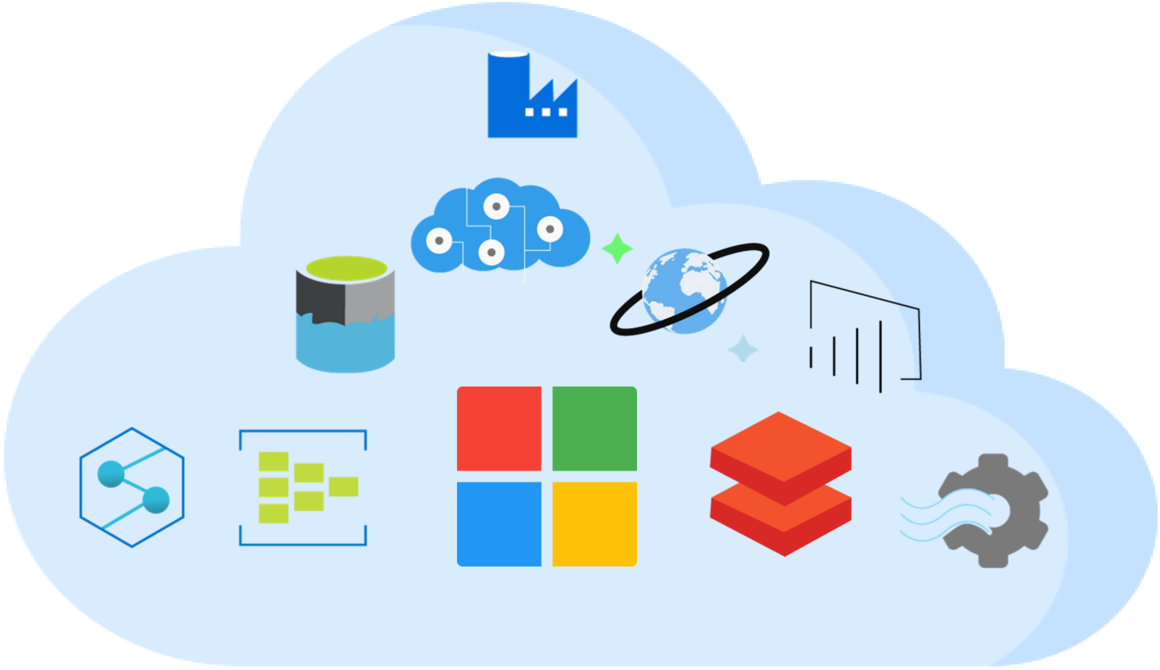
Recommend a Data Flow to Meet Business Requirements

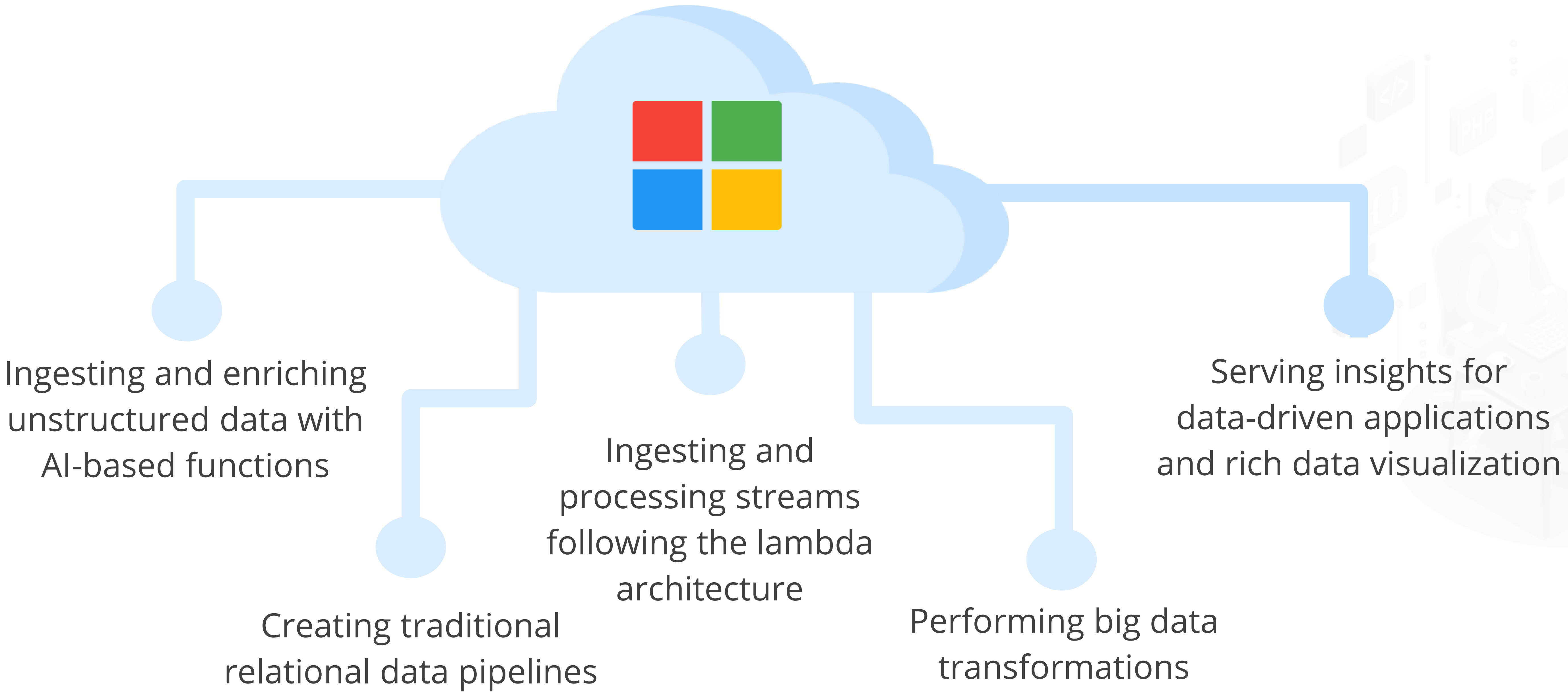
Azure Data Services

The extensive family of Azure Data Services can build a modern data platform for handling the most common organizational data challenges.



Azure Data Services

This solution architecture demonstrates how you can use a single, unified data platform to meet the most common requirements for:



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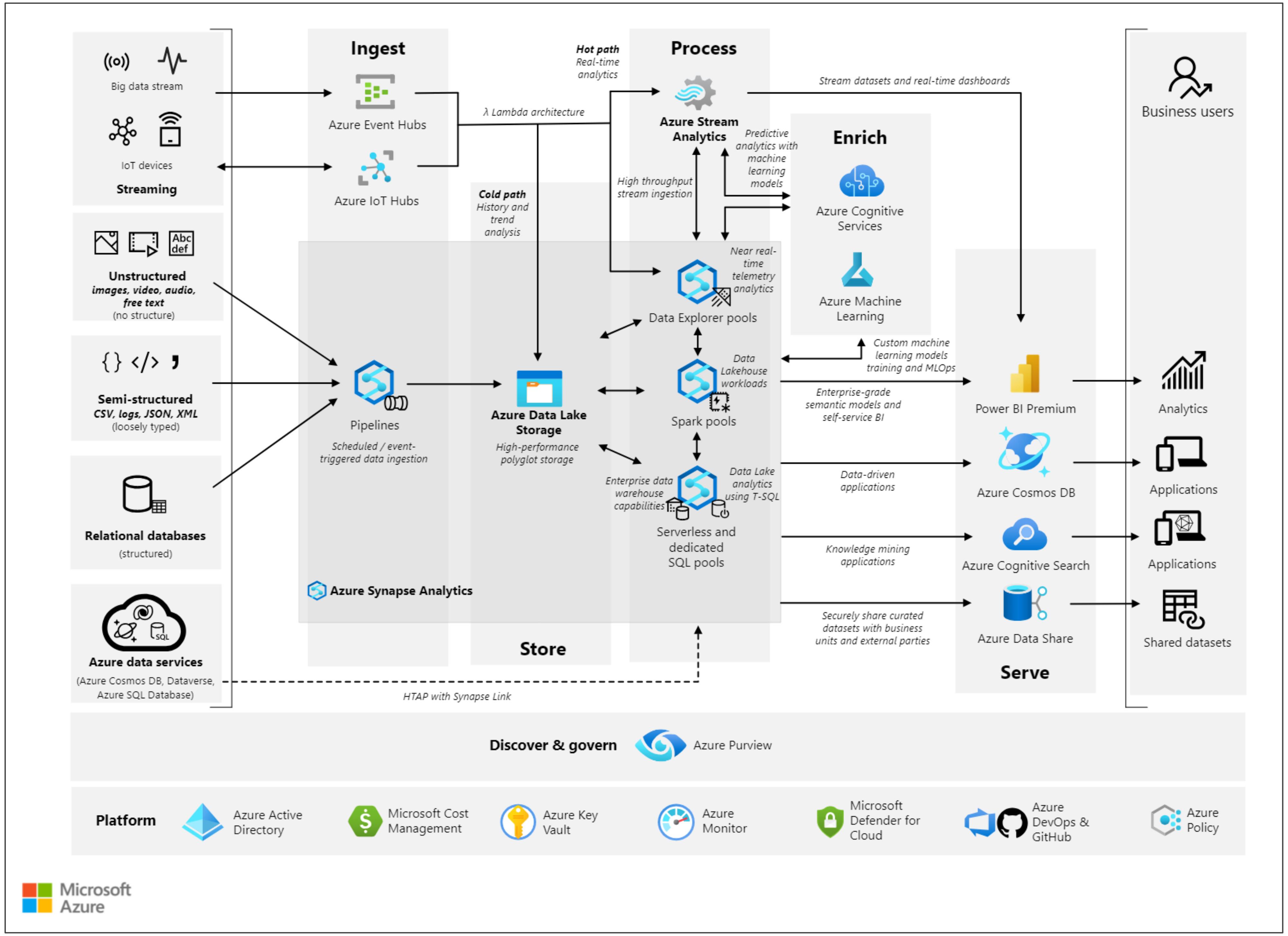
Azure Data Services: Reference Architecture



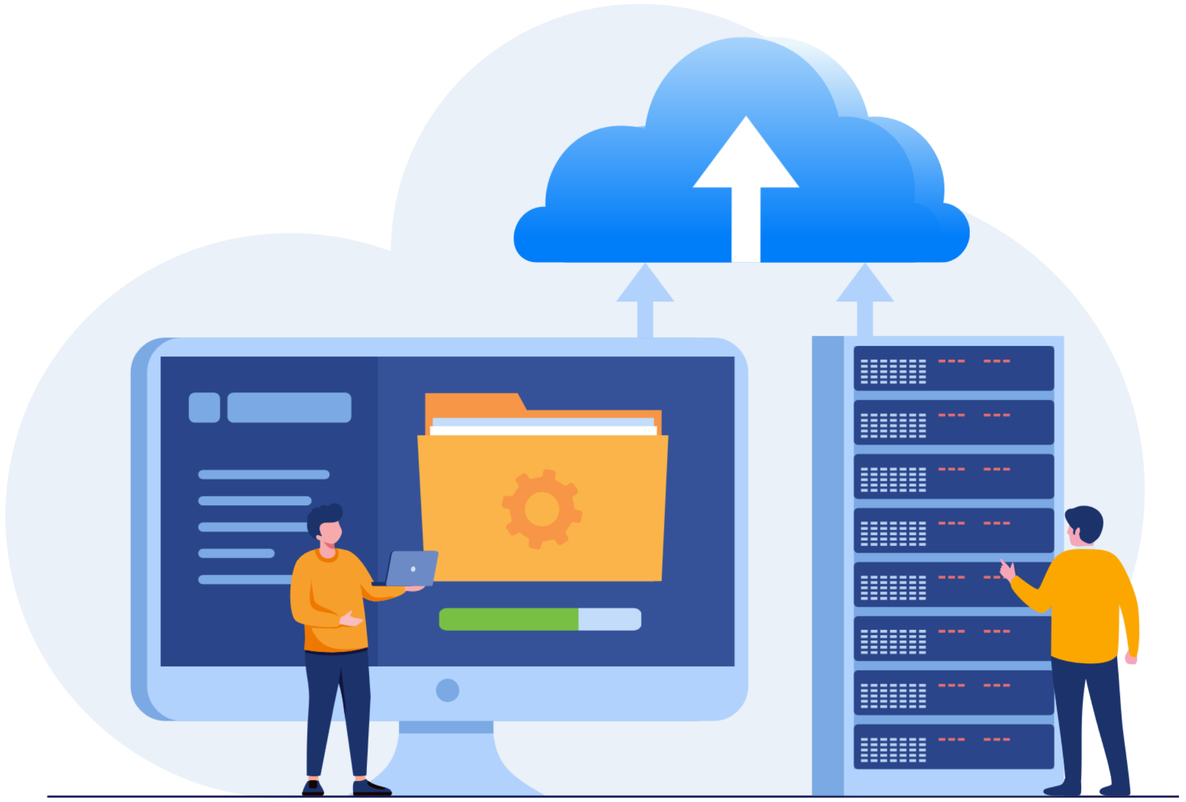
Azure Data Services

It provides the following services:

- Azure Data Factory
- Azure Synapse Analytics
- Azure Cosmos DB
- Azure Event Hubs
- Microsoft Power BI
- Azure Data Lake Gen2
- Azure Databricks
- Azure Cognitive Services
- Azure Stream Analytics

Azure Data Lake Storage

It is a scalable and secure data lake solution for big data analytics.



It is designed to handle massive amounts of data with high-throughput for analytics and machine learning.

Azure Data Lake Storage

The key features include:

Scaling effortlessly to handle petabytes of data, accommodating the growing needs of big data applications

Integrating seamlessly with Azure services, including Azure Databricks, Azure Synapse Analytics, and so on

Ensuring data security and compliance through granular access controls, encryption, and Azure AD integration

Azure Data Services

The use cases include:

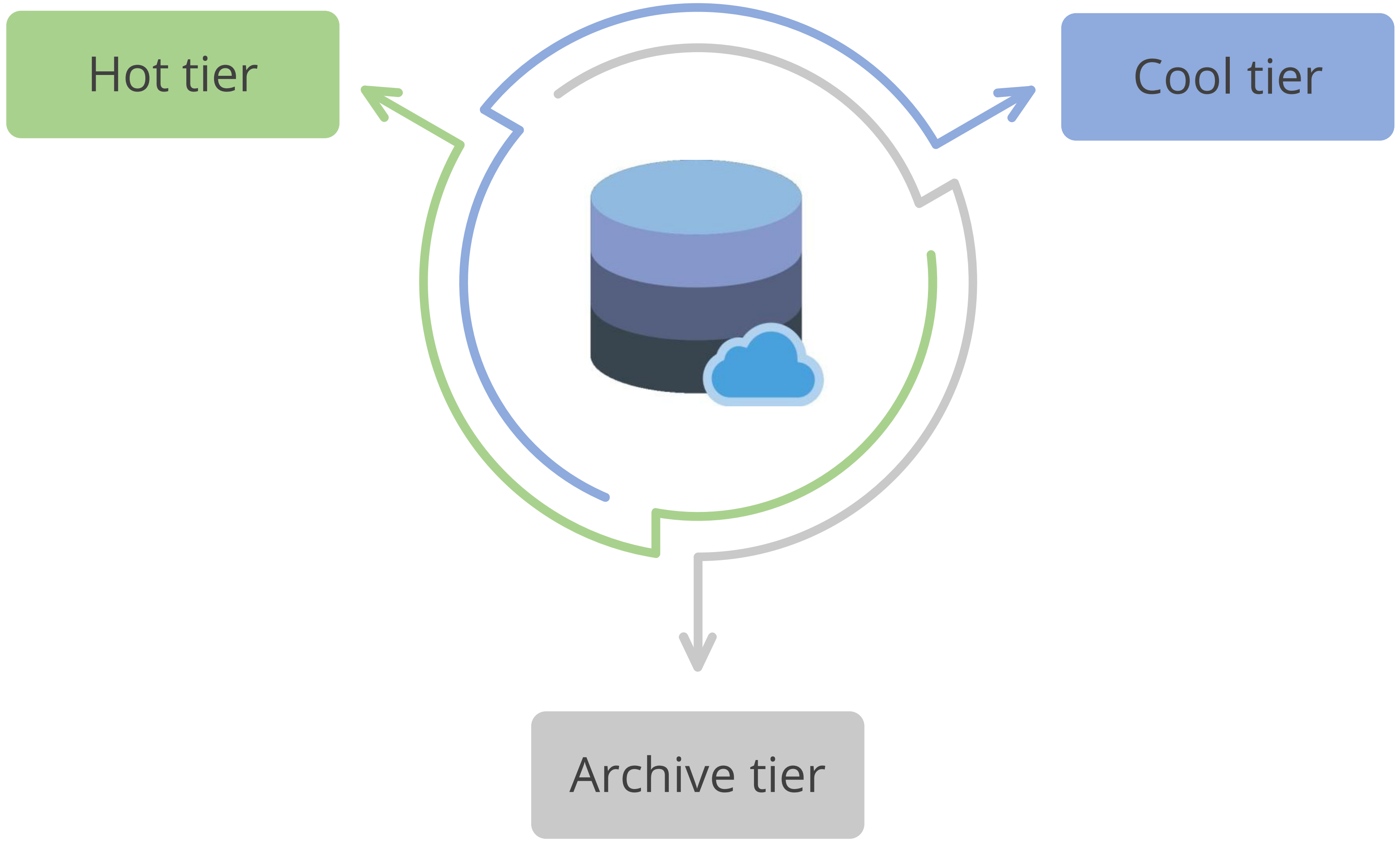
Establish an enterprise-wide data hub encompassing a structured data warehouse and a semi-structured and unstructured data lake

Employ big data processing techniques to integrate relational data sources seamlessly with other unstructured datasets

Use semantic modeling and advanced visualization tools to enhance data analysis capabilities

Azure Data Lake Storage

Storage tiers include:



Azure Data Lake Storage

The use cases include:

Big data analytics	Data lakes
Ideal for processing and analyzing large datasets to derive insights and support decision-making	Enables the creation of scalable and secure data lakes for storing structured and unstructured data

Azure Data Lake Storage

Best practices:

Design effective data partitioning to optimize query performance

Implement RBAC to control access at different levels

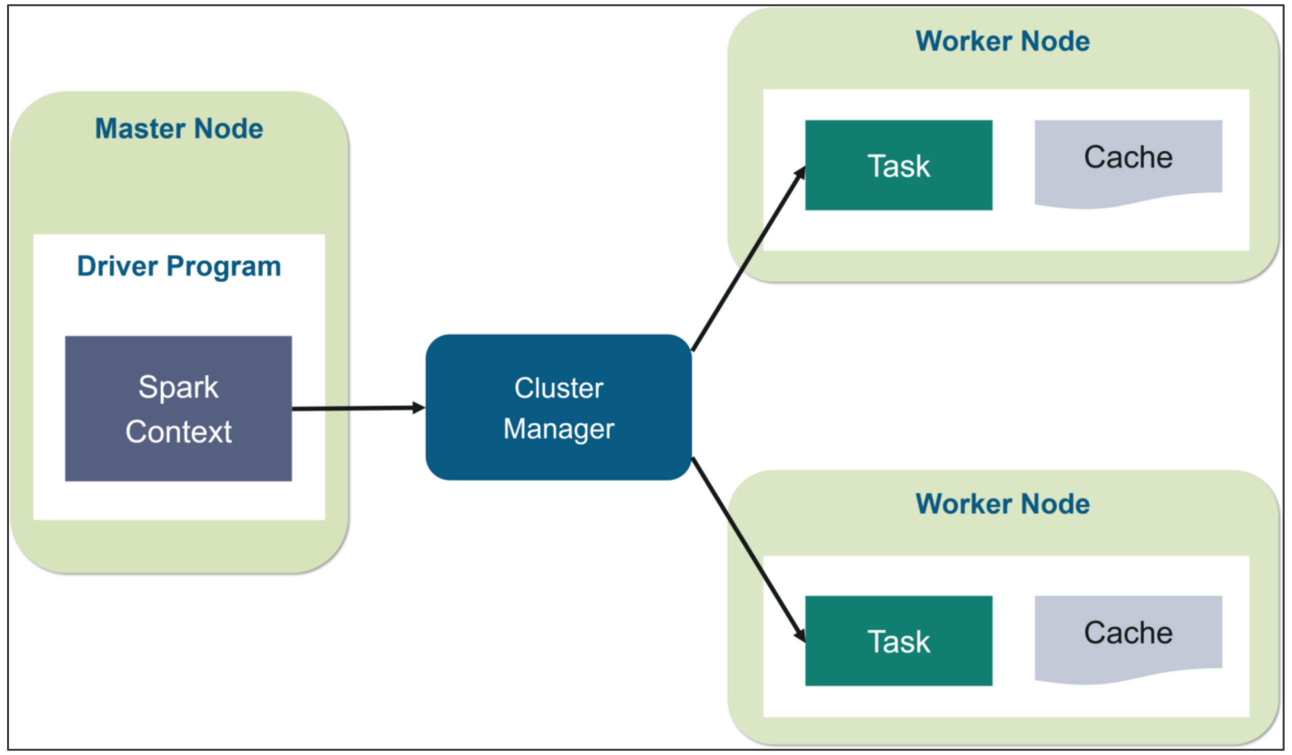
Azure Databricks

It provides the latest versions of Apache Spark and allows users to integrate with open-source libraries.

It accelerates big data analytics and AI solutions with a collaborative and scalable environment. It supports multiple languages, including Python, Scala, SQL, and R.

Apache Spark

It works by distributing data and computational tasks across multiple nodes in a cluster, enabling parallel processing of large datasets in memory.



- Databricks is designed to simplify the deployment, management, and scaling of Apache Spark clusters in the cloud.
- It offers a collaborative platform for data engineering, data science, and machine learning, providing optimized performance, streamlined workflows, and seamless integration with cloud storage and computing resources.

Azure Databricks

Best practices:

Optimize cluster configurations for cost and performance

Follow best practices for managing workspaces, notebooks, and libraries

Azure Databricks

The use cases include:

Big data analytics	Data engineering	Machine learning
Efficiently processes and analyzes large volumes of data	Streamlines ETL processes and data engineering tasks	Supports the development and deployment of machine learning models at scale

TECHNOLOGY

Recommend a Solution for Data Integration

Azure Data Factory

It easily constructs ETL (extract, transform, and load) and ELT (extract, load, and transform) processes code-free.

Ingest

- Enables multi-cloud, on-premise, and hybrid data copying
- Leverages 90+ native connectors for diverse data sources
- Utilizes serverless and auto-scaling capabilities

Control flow

- Designs code-free data pipelines for seamless orchestration
- Generates pipelines programmatically via SDK
- Uses workflow constructs such as loops and branches

Azure Data Factory

Data flow

- Implement code-free data transformations using Spark
- Scale-out capabilities with Azure integration runtime

Schedule

- Builds operational schedules for the data pipelines
- Maintains event-based, tumbling, and chained windows

Monitor

- Views active executions and pipeline history
- Establishes alerts and notifications

How Data Factory Works?

It contains a series of interconnected systems that provide a complete end-to-end platform for data engineers.

Connect and collect

Data Factory facilitates data movement from on-premises and cloud source data stores to a centralized data store for analysis using the **Copy activity** in a data pipeline.

Transform and enrich

Data flows empower data engineers to create multiple data transformation graphs that run seamlessly on Spark, eliminating the need for in-depth knowledge of Spark clusters or programming.

How Data Factory Works?

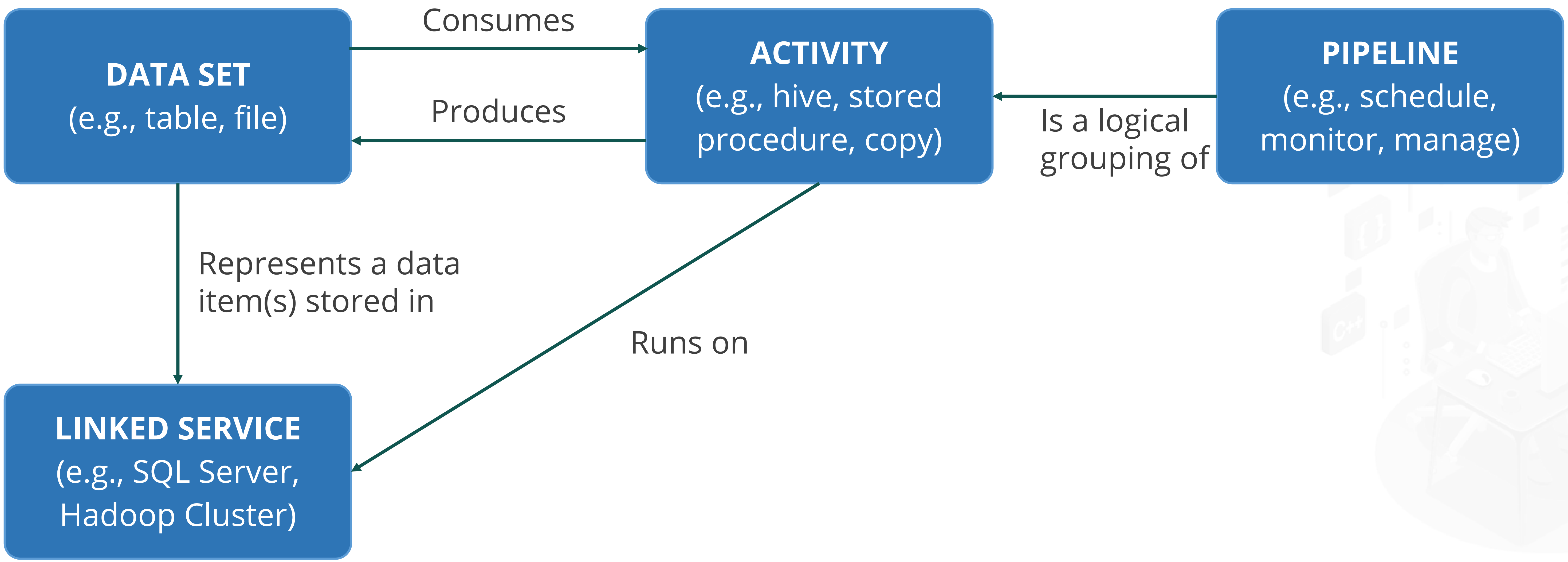
CI/CD and publish

Full support for CI/CD of data pipelines using Azure DevOps and GitHub enables incremental development and ETL procedure delivery before publishing.

Monitor

Pipeline monitoring is built-in with Azure Monitor, API, PowerShell, Azure Monitor logs, and Azure portal health panels.

Working of Azure Data Factory Components

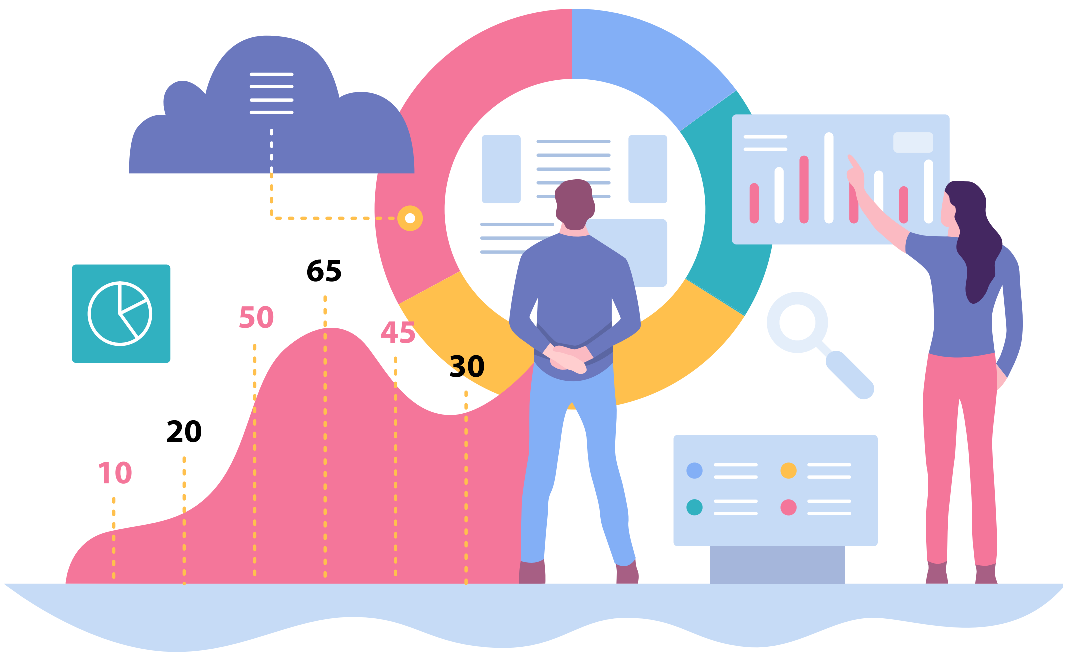


Key Components of Azure Data Factory

Datasets	They are data structures within data stores that refer to the data you want to use as inputs or outputs in their operations.
Activities	They represent a processing step in a pipeline. For example, you could use a copy activity to transfer data from one data store to another.
Pipelines	They are a logical grouping of activities that perform a unit of work.
Linked service	It establishes a connection to the data source; a dataset describes the data's structure.

Azure Synapse Analytics

It is an analytics service that joins enterprise data warehousing and big data analytics with a unified experience to ingest, prepare, manage, and serve data for immediate BI and machine learning needs.

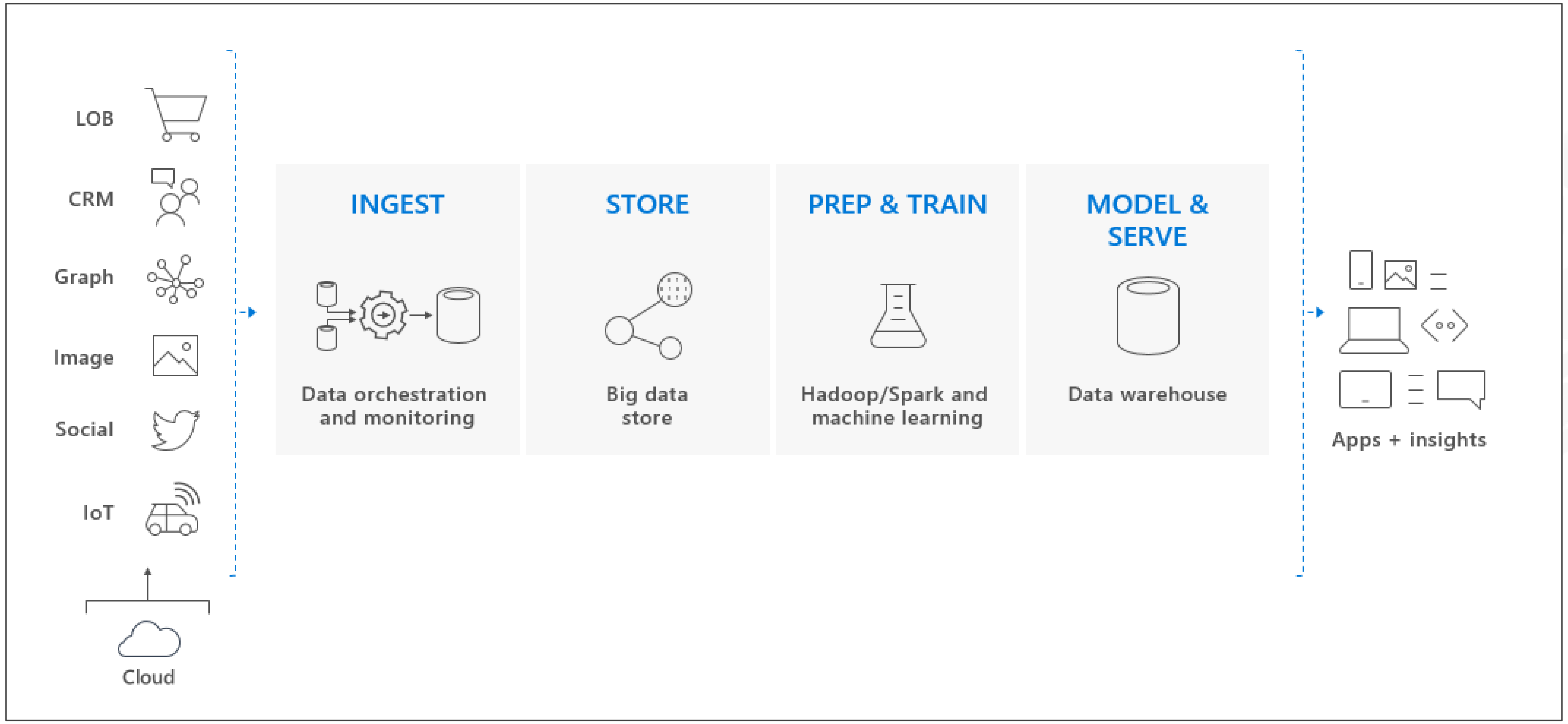


Azure Synapse include four components:

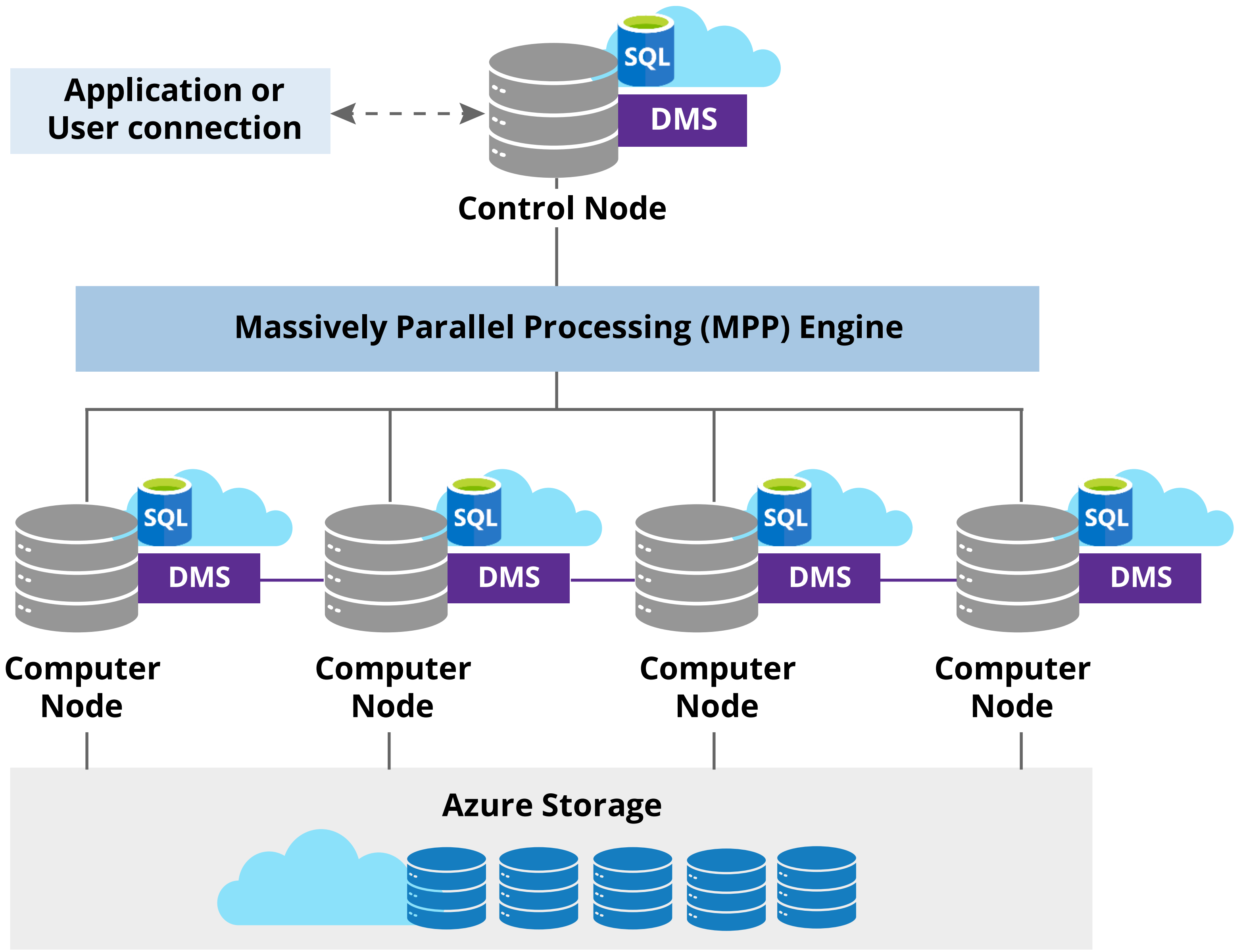
- Synapse SQL
- Spark
- Synapse Pipelines
- Studio

Dedicated SQL Pool

It refers to the enterprise data warehousing features available in Azure Synapse Analytics.

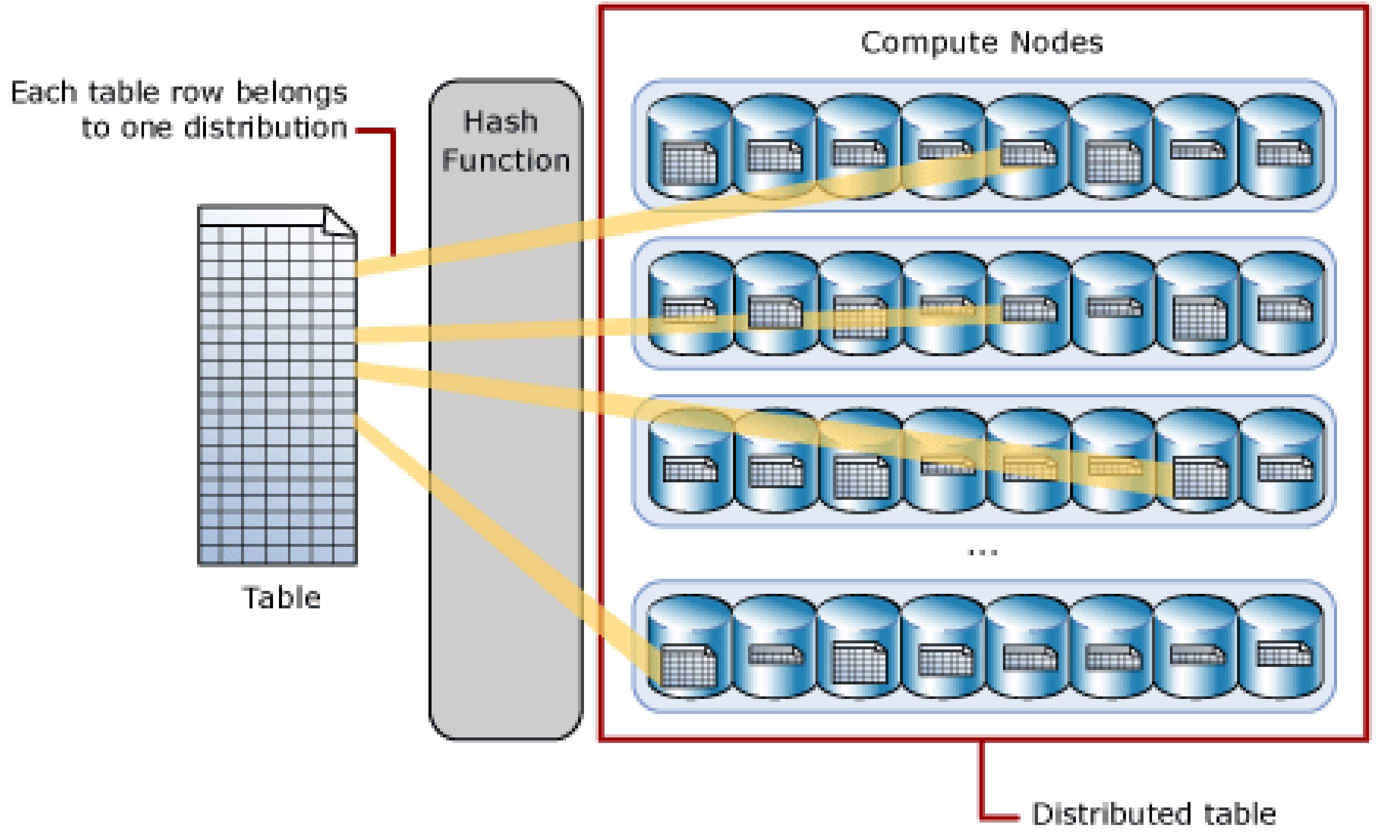


Azure Synapse SQL Architecture



Distributions

A distribution is the basic unit of storage and processing for parallel queries that run on distributed data.



Hash-distributed tables

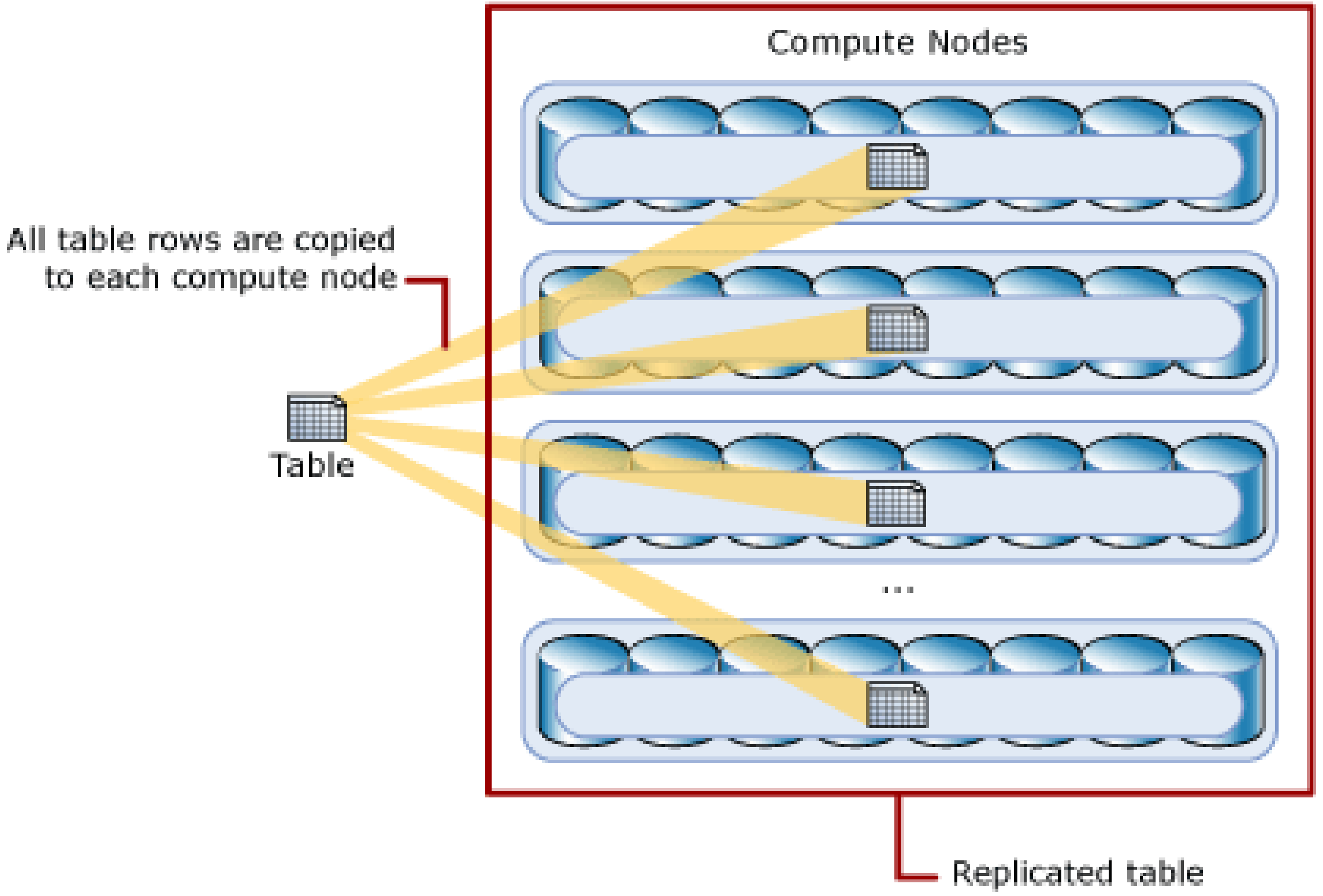
They deliver the highest query performance for joins and aggregations on large tables.

Round-robin distributed tables

They deliver fast performance when used as a staging table for loads.

Replicated Tables

It provides the fastest query performance for small tables.



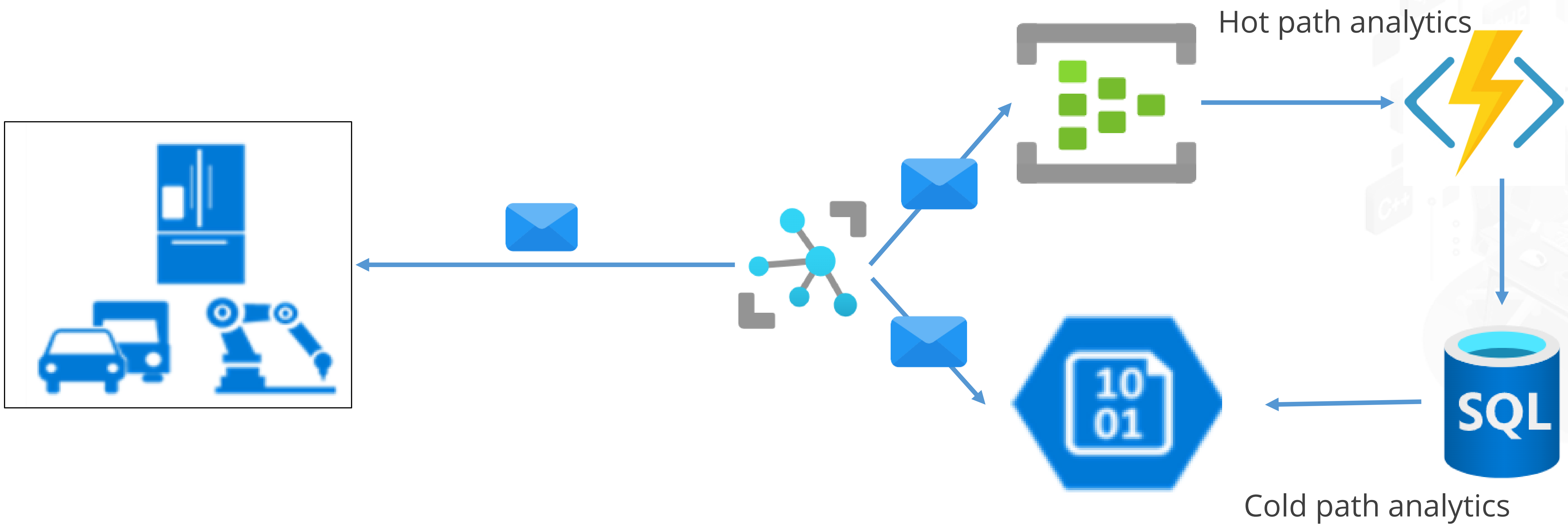
- A replicated table caches a full copy of the table on each compute node.
- It removes the need to transfer data among compute nodes.
- It is best utilized with small tables.

TECHNOLOGY

Design a Strategy for Hot, Warm, and Cold Data Path

Working with Azure IoT Data

There are some patterns in how IoT use cases consume, analyze, and store data. The following diagram depicts an IoT pipeline in which IoT devices deliver data to Azure IoT Hub and then to Azure SQL Database:



Hot Path

It is dedicated to real-time data processing and display, supporting real-time alerting and streaming activities.

- Leverages an Azure Function App, SignalR, and a hosted web app for immediate alerts and data streaming
- Enables streaming data through a WebSocket-based connection via Azure SignalR and the web

Warm Path

It stores or displays only a recent subset of data.

- Within the warm path, minor analytic and batch processing procedures are executed to retain or display only the latest segment of the data.
- This procedure involves utilizing a web app built on an Azure App Service.

Cold Path

It is used for long-term data storage.

- The cold path involves extensive analytics and batch processing reserved for long-term data storage.
- Azure data explorer is used here because it saves data efficiently for long periods, presently with a default of 100 years.

Scenarios to Use Hot, Warm, and Cold Path

Hot data path:

- A smart factory where sensor data is continuously monitored to detect equipment malfunctions in real-time, use a hot data path.
- IoT-enabled health monitoring systems that track vital signs in real-time to alert healthcare providers of critical changes.

Warm data path:

- A smart building system that collects temperature, humidity, and occupancy data for hourly or daily analysis to optimize energy consumption, use a cold data path.
- Fleet management systems that analyze data daily or weekly to optimize routes and fuel usage.

Scenarios to Use Hot, Warm, and Cold Path

Cold data path:

- A logistics company storing vehicle telemetry data from the past 5 years to analyze long term trends or for compliance purposes.
- Storing historical environmental sensor data for regulatory reporting or future analytics on climate patterns.

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Design Azure Stream Analytics Solution for Data Analysis

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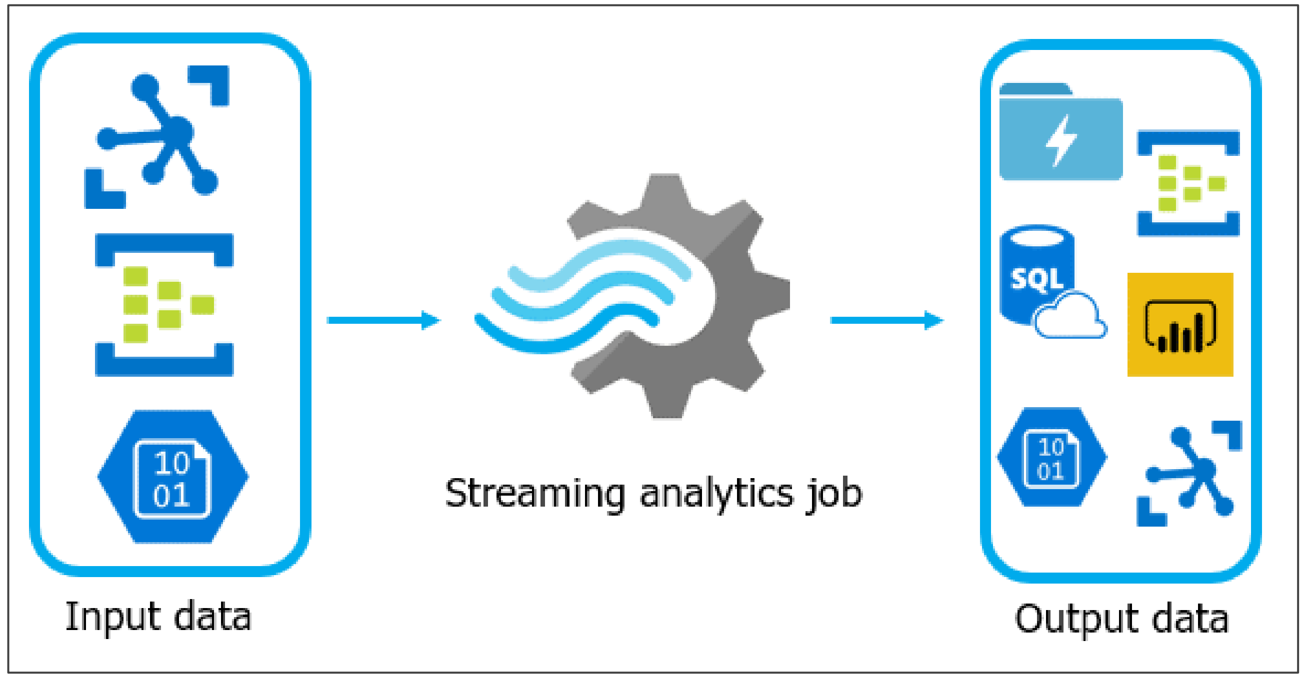
Azure Stream Analytics

It is a fully managed, serverless, real-time analytics engine.



It is designed to analyze and process large volumes of streaming data with sub-millisecond latencies.

Azure Stream Analytics

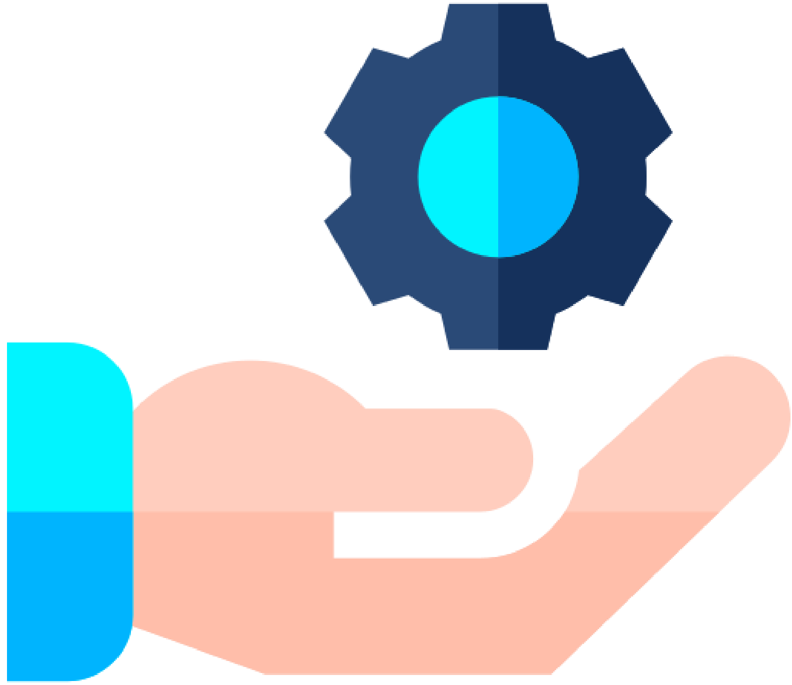
The following are the use cases of Azure Stream Analytics:

- Dashboarding in real-time using Power BI for monitoring purposes
- Storing streaming data for later access by other cloud services
- Transforming and analyzing data in real-time to trigger processes
- Sending alerts
- Making judgments in real-time
- Working in machine learning with limited application in more complex analytics
- Operating in blob storage accounts

Azure Stream Analytics: Features

- Sends data to services like Azure Functions, Service Bus topics, or queues to trigger communications or custom actions further down the line
- Sends data to a Power BI dashboard for real-time dashboarding
- Stores data in other Azure storage services (for example, Azure Data Lake and Azure Synapse Analytics) to train a machine learning model or execute batch analytics based on historical data

Benefits of Azure Stream Analytics



- Ease of use
- Ease of adoption
- Fully managed
- Reliability
- Security
- High performance

Limitations of Azure Stream Analytics

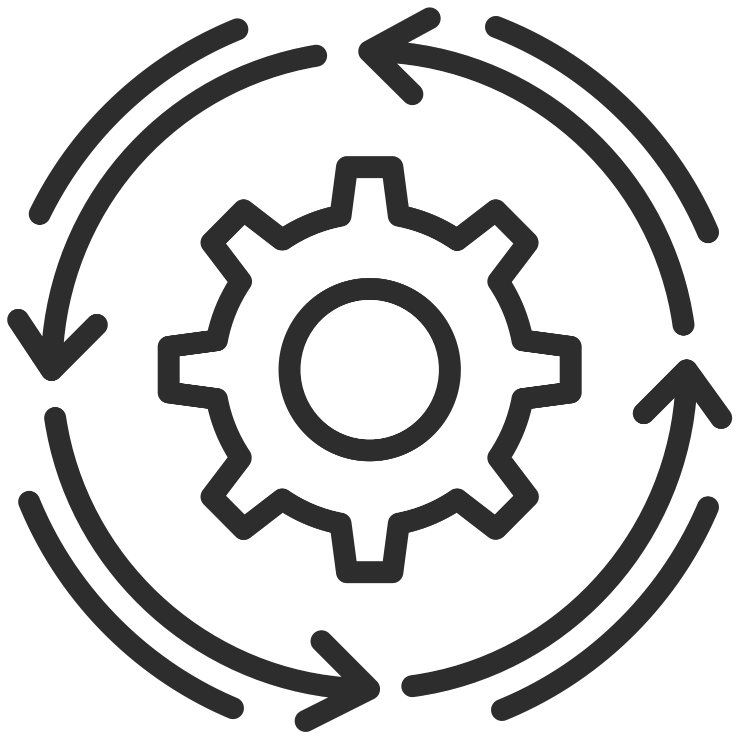
- It only works with SQL.
- The input data must be in AVRO, JSON, or CSV format.
- Blob storage can only be used to store static data.
- Integration is limited to Azure services.
- Users cannot leverage dynamic reference data join functionality.
- Automated scaling (scaling jobs in Azure Portal) is not available.

Azure Stream Analytics Job

The no-code editor allows you to develop a Stream Analytics job effortlessly to process real-time streaming data, using drag-and-drop functionality, without writing a single line of code.

It comprises three main components: streaming inputs, transformations, and outputs.

Azure Stream Analytics



Workflow:

- An Azure Stream Analytics job is composed of three components: an input, a query, and an output.
- The job retrieves data from Azure Event Hubs, Azure IoT Hubs, and Azure Blob Storage.
- The query is written in a SQL-like language, allowing users to easily filter, sort, aggregate, and merge streaming data.
- Additionally, users can reference a workflow within another workflow for more complex processing.

High Level Approach by Data Analytics

Data is ingested into the Azure Streaming Platform from IoT devices, log files, Azure Blob, and financial data such as credit card transactions.

Users can use a Stream Analytics query to perform data analytics, including sorting, filtering, and aggregations.

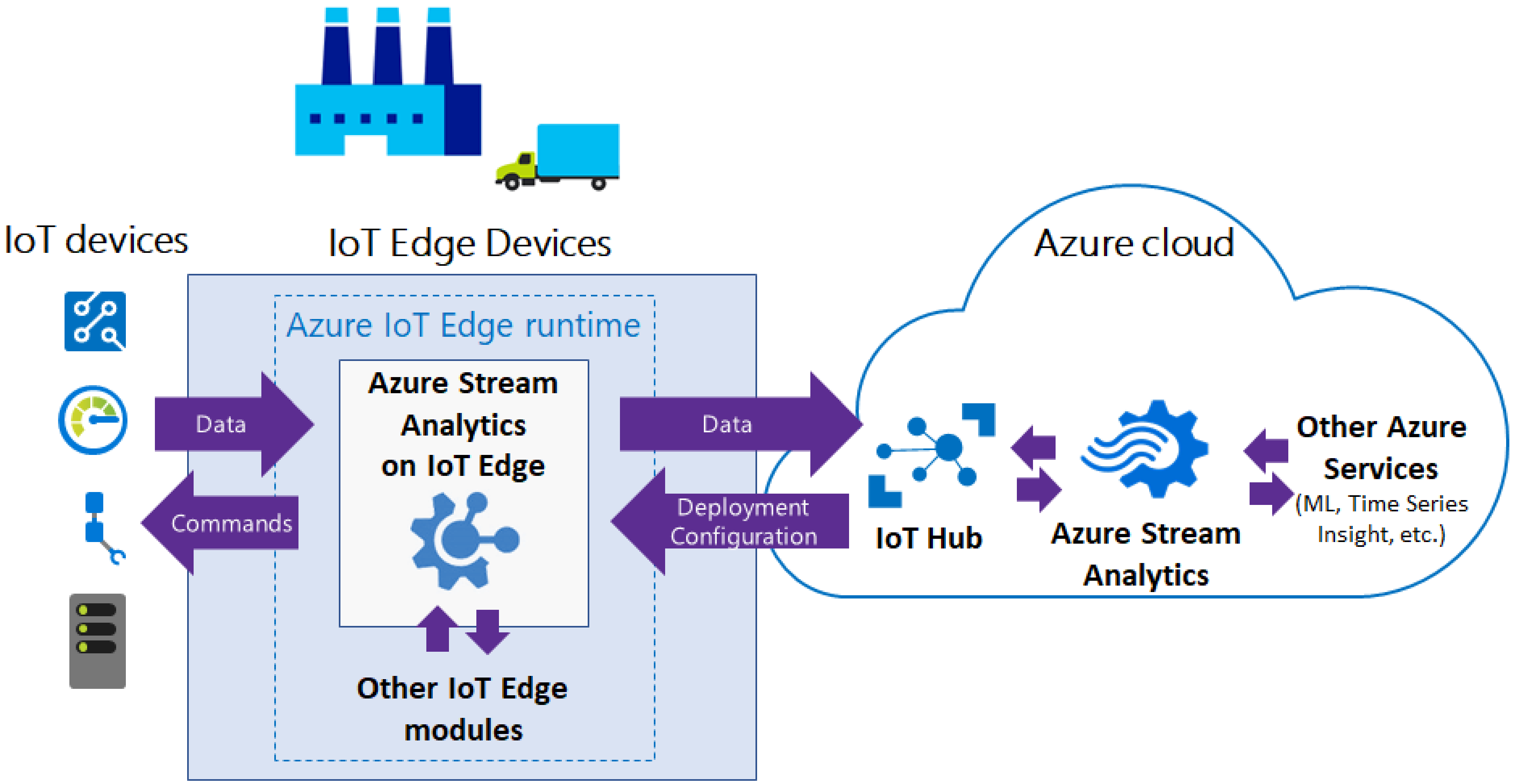
Downstream applications, such as data warehouses and reporting tools, can leverage the processed data for additional analysis and issue alerts and notifications.

Azure Stream Analytics Pricing Structure

Parameters	Standard	Dedicated
Resource type	Stream Analytics Job	Stream Analytics Cluster
Minimum unit required	1/3 Streaming Unit (fraction of a node)	12 Streaming Units
Virtual network support	Yes	Yes
Built-in customized functions	Yes	Yes

Azure Stream Analytics on IoT Edge

It empowers developers to deploy near-real-time analytical intelligence closer to IoT devices to unlock the full value of device-generated data.



Key Takeaways

- Azure Data Services builds a modern data platform that can handle the most common organizational data challenges.
- Azure Data Factory easily constructs ETL (extract, transform, and load) and ELT (extract, load, and transform) processes code-free.
- Azure Synapse Analytics joins enterprise data warehousing and big data analytics with a unified experience.
- Azure Stream Analytics is a fully managed, serverless, real-time analytics engine.

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Knowledge Check

Knowledge Check

1

Which of the following meets the requirements for a unified data platform?

A. Rich data visualization

B. Relational data pipelines

C. Big data transformations

D. Non-stream ingestion

Knowledge Check 1	Which of the following meets the requirements for a unified data platform?

A. Rich data visualization

B. Relational data pipelines

C. Big data transformations

D. Non-stream ingestion

The correct answers are **A, B, and C**

A unified data platform combines rich data visualization, supports relational data pipelines, and handles big data transformations for versatile and effective data processing.

Knowledge Check 2	**Which of the following Azure services provide a tool for big data analytics, business intelligence, and machine learning?**

A. Data factory

B. Cognitive services

C. Synapse analytics

D. Databricks

Knowledge Check

2

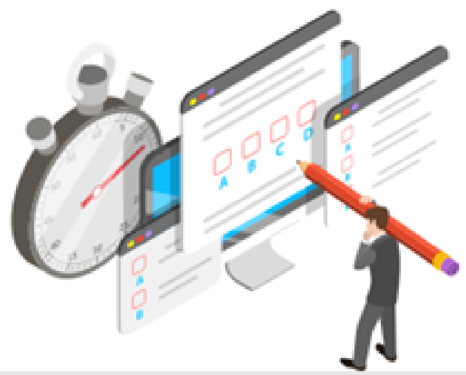
Which of the following Azure services provide a tool for big data analytics, business intelligence, and machine learning?

A. Data factory

B. Cognitive services

C. Synapse analytics

D. Databricks



The correct answer is **C**

Azure Synapse is an analytics service that combines enterprise data warehousing and big data analytics, serving data for quick business intelligence and machine learning needs.

Knowledge Check
3

Which of the following terms is used as a fundamental unit of storage and processing for parallel queries on distributed data?

A. Distribution

B. Data movement

C. Compute node

D. Control node

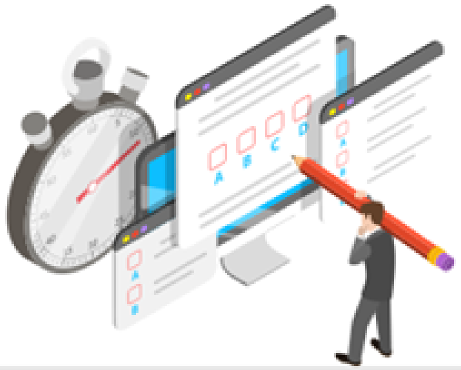
Knowledge Check 3	Which of the following terms is used as a fundamental unit of storage and processing for parallel queries on distributed data?

A. Distribution

B. Data movement

C. Compute node

D. Control node



The correct answer is **A**

Distribution is the basic storage and processing unit for parallel queries that run on distributed data.

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Thank You